

The relevance of features to limit order book learning

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Abstract

The promising predictive power of machine learning (ML) models is encouraging a wide range of applications, but the weak interpretability of their black-box feature is a major obstacle to their application to finance. The large scale of limit order book (LOB) data provides fertile ground for ML implementation and exploration in finance. To examine the contribution of input elements to ML models, we propose three novel statistical measures for ranking each feature set from the LOB data and the combinations of feature sets: the p -value from hypothesis testing with a black-box model, a newly defined out-of-sample R^2 with respect to the objective function, and another new R^2 with respect to the classification score. Multiple measures are used to evaluate the level of significance of the features and their consistency. We empirically test several distinct models using LOB data from the Chinese A-share market. We find that feature relevance varies greatly across models for LOB forecasting. Moreover, notable patterns of consistency and significance offer important insights into the remarkable empirical performance of some black-box models. The testing results motivate us to improve prediction with ensemble learning. An additional sensitivity analysis clarifies that the feature rankings are robust to off-machine parameters under the well-designed deep convolutional neural network.

Keywords: Machine learning; Explainable methods for black box; limit order book; deep learning; Chinese A-share market

1. Introduction

Although data-driven models powered by machine learning (ML) techniques are shown to be useful in a wide range of applications, the black-box features of these models mean that it is usually unclear how they derive a certain decision, and therefore, explanations of their prediction accuracy and stability are lacking. Traditional statistical methods are inadequate to carry out structural and technical analyses to understand the rationale underlying the predictions from black-box models. Consequently, explainable artificial intelligence (XAI), which aims to elucidate why and under what conditions a model arrives at a specific decision, is attracting a great deal of attention. XAI enables human beings to understand data-driven ML algorithms and detect erroneous reasoning (Lapuschkin et al., 2019; Lundberg et al., 2020). In the credit risk area, this enables the predictions of default probability and loss given default by black-box models to be explained in terms of their inputs (Bussmann et al., 2021; Bastos & Matos, 2022; Kellner et al., 2022), which is particularly important because transparency is required to comply with the Basel II agreement. In this study, we propose a set of measures to explain the performance differences between black- and white-box models, and we apply them to empirically interpret the forecasts of limit order book (LOB) learners.

After years of ML development, numerous publications are dedicated to the application of black-box models in stock markets, as the inherent noise and stochastic nature of these markets limit the utilization of traditional white-box models. For interday trading, Leippold et al.

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(2022) and Peng et al. (2021) predict stock returns based on a comprehensive set of factors using various ML algorithms. In the case of intraday trading, the high-frequency financial data make stock price patterns more sophisticated and volatile than interday patterns. To capture the essentials, ML models, such as support vector machines (SVM) (Kercheval & Zhang, 2015), multilayer perceptrons (MLP) (Nousi et al., 2019), and deep neural networks (Chong et al., 2017; Zhang et al., 2019), are used to contribute to forecasting short-term price movements of stocks and achieve greater out-of-sample precision than simple linear methods (Briola et al., 2020). The LOB data and elicited features are well utilized in the literature, but proper in-depth explanations of the use of these features is lacking. This may undermine confidence in a wide range of applications and hinder future improvements.

Several methods or measures for uncovering black boxes are proposed in the literature. Shapley values (Lundberg & Lee, 2017) are designed to measure the importance of variables, but their calculation is computationally demanding for high dimensions. Another option is an accumulated local effects plot (Apley & Zhu, 2020), which is a visual representation of how a single predictor influences the model predictions; however, it is not workable for multiple predictors. Model-based significance tests by Horel & Giesecke (2020) discern the impact of individual variables but are restricted to a single-layer feed-forward neural network model.

In this paper, we propose a feature explanation framework with three complementary statistical measures for LOB learning models. The first measure is based on model-free hypothesis testing with unknown limiting distributions of parameter estimates. Specifically, we adopt the statistical significance test derived by Dai et al. (2022) as a simultaneous test of a collection of features that does not require massive computing. That is, we attempt to explain the feature relevance for the intractable LOB forecasting algorithms using model-free p -values. The novel p -values have the same statistical interpretation as classical p -values. Loosely speaking, the p -value of an input feature reflects the likelihood that the contribution of this feature to prediction is purely by chance under the null hypothesis. Thus, the lower (higher) the p -value, the higher (lower) is the significance level of an input feature. The other two measures that we propose for evaluating feature importance are novel out-of-sample R-square measures with respect to the predictive error ($\tilde{R}_{\text{error}}^2$) and the area-under-the-curve (AUC) score ($\tilde{R}_{\text{AUC}}^2$). More convincing and compelling results can be achieved by using multiple measures to explore the facts than using one measure only (Huynh-Thu et al., 2012). These three measures not only enable us to test feature significance under black-box models but also provide a method to compare traditional and state-of-the-art models in terms of measure consistency. More precisely, a stable and consistent learning framework is expected to produce a consistent ranking for feature selection among the three measures. Our empirical study finds that a deep convolutional neural network (DCNN) stands out in LOB learning in terms of these three measures. It also performs best in terms of the conventional Sharpe ratio for profit performance.

To better understand the nonlinear mapping between the input features and output predictions of LOB models, we test for a collection of hypothesized features. In Section 2, four major feature sets are used: raw LOB data, order flow, a time-insensitive set, and a time-sensitive set. These feature sets describe the market from different perspectives, serve as distinct functional regions, and contribute insights into the short-term price trend to varying degrees. They are fed into both white- and black-box models, including multinomial logistic regression, multi-class SVM, random forest, and DCNN models, as we discuss in Section 3. The models extract compressed and invisible knowledge from the features and then predict the nature of trading signals as *long*, *short*, or *none*. Then, the obtained signals are put forward to generate investment strategies. Under the complete scheme, our objective is to investigate the roles of both individual and integrated feature sets for LOB prediction.

We conduct an empirical study using Chinese A-share market data to test the marginal effect of each feature set and the joint importance of combined feature sets. The implementation of the three explainable measures reveals the information on which the predicted signals are

based. First, the most important determinants of trading signals differ for all of the models under investigation, implying that the importance of the variables is sensitive to the model structure. Second, we determine and analyze the consistency between the three measures and the significance of the features within the same model. For DCNN, the three measures present consistent rankings for 14 features with a certain degree of significance. Conversely, for the other three models, no consistency is found for the logistic regression, the significance levels are generally very low for SVM, and the random forest method leads to performances exhibiting great disparities. Furthermore, the patterns and behaviors of the feature rankings help explain the performance of the original models. Both the measure consistency and the feature significance support the conclusion that the DCNN has a stronger ability than the other measures to digest features and capture valuable internal connections because of its complex nonlinear structure.

To the best of our knowledge, this paper is the first to articulate and explain the remarkable performance of deep learning for LOB forecasting and trading. The inconsistency and insignificance of input features found when using the parametric linear models of logistic regression and SVM show that a successful LOB prediction relies on a complex and nonlinear transformation of the input features. Further, we empirically confirm that the nonlinear ML framework of the random forest successfully identifies several significant input features, each of which consistently improves prediction. However, the random forest does not fully utilize the nonlinear information contained in the rest of the features. The DCNN gains predictive improvement from all of the input features, as reflected in the prediction measures of $\tilde{R}_{\text{error}}^2$ and $\tilde{R}_{\text{AUC}}^2$. In addition, the significance levels based on the p -values are consistent with the prediction measures for almost all of the input features. Ranking the significance of the features helps us to understand the key decision variables of a black-box model if there is consistency between the three measures. Although parametric models are easy to interpret, the conclusions developed from a parametric model could be in doubt if the model is consistently inconsistent in terms of its significance and prediction analysis.

The learned experience from black-box models is crucial and informative for improving models. We provide a preliminary demonstration. Considering the results for feature relevance tested on 30 stocks in the Chinese A-share market, we adopt ensemble learning with the bagging algorithm to combine the DCNN and random forest. First, we train the two models independently on the most significant feature sets and then use the weighted combined model for predictions. We find that both prediction accuracy and actual profit are promoted when the models are combined, indicating that input features are well-pruned and utilized. On the one hand, a single simplified model with fewer features contains less disturbance and noise than a combined model, and the feature selection procedure is beneficial. On the other hand, subsequent ensemble modeling can incorporate refined intelligence that originates from different feature sets altogether. It ensures that feature sets not selected by one model are processed in another model and that no related information is overlooked.

Finally, we conduct an additional sensitivity analysis to show the robustness of our measures under the DCNN. Consistent rankings of feature importance for each experiment indicate that the time period, data split, and stock selection are not substantial off-machine parameters, instead being trivial external settings that do not contribute greatly to the feature relevance rankings. Therefore, our interpretations hold true for universal stocks in the market and are not limited to short time periods. This sensitivity analysis strengthens our confidence in the model-free significance test and supports the adoption of these measures in a wide range of areas.

The rest of this paper is organized as follows. In Section 2, we introduce the concepts and definitions of the four major feature sets. Section 3 describes the four distinct models for forecasting the trading signals in detail. We explain and conduct the significance tests of the feature sets using the three proposed measures (p -value, $\tilde{R}_{\text{error}}^2$, and $\tilde{R}_{\text{AUC}}^2$) in Section 4. Then,

in Section 5, we empirically evaluate the performance of the feature testing and improve the predictions for the Chinese A-share market. A sensitivity analysis is also included. Section 6 concludes the paper. The technical details are provided in the Appendix.

2. Input feature sets

As data-driven algorithms, ML frameworks for LOB require essential “ingredients” as their foundation. These ingredients are different forms of data and features. In addition to raw LOB data (Zhang et al., 2019; Yin & Wong, 2022), many features are proposed and tested in the literature. Tsantekidis et al. (2020) demonstrate that their proposed stationary features significantly outperform the raw price features, and Kolm et al. (2021) show that models trained on order flow significantly outperform those trained directly on order books. Many other handcrafted features are designed and evaluated by Kercheval & Zhang (2015); Ntakaris et al. (2019); Nousi et al. (2019). Our choice of the feature sets that follow is based on their practical implications, tractable mathematical calculations, and experimental performance in the literature.

2.1. Raw limit order book data

We first introduce the direct and visible high-frequency data sent by the exchanges at every tick time t . The basic features of the tick-by-tick LOB data are prices and volumes at each level from both the ask and bid sides of the market. Let

$$\mathbf{x}_{1t} := \left[p_a^{(i)}(t), v_a^{(i)}(t), p_b^{(i)}(t), v_b^{(i)}(t) \right]_{i=1}^n \in \mathbb{R}^{4n}, \quad (2.1)$$

where $p_a^{(i)}(t)$ and $v_a^{(i)}(t)$ denote the ask price and ask volume, respectively, at level i at time t . Similarly, $p_b^{(i)}(t)$ and $v_b^{(i)}(t)$ denote the price and volume, respectively, on the bid side.

The data describe the shape of markets and summarize the fundamental trading circumstances. Listed orders with prices and volumes contain a great deal of valuable content. However, inherent noise and randomness are inevitable and will distort predictive power. Therefore, we derive additional feature sets from the raw LOB data with the aim of extracting more granular and refined information.

2.2. Order flow

To capture the transformation of the LOB, we integrate information on volumes and prices. Simply put, this feature set is derived by calculating changes of volumes conditioned on the price changes at each level. We use the following definition of order flow at tick time t as the vector:

$$\mathbf{x}_{2t} := \left[\text{bOF}^{(i)}(t), \text{aOF}^{(i)}(t) \right]_{i=1}^n \in \mathbb{R}^{2n}, \quad (2.2)$$

where the bid order flows (bOF) and ask order flows (aOF) proposed by Kolm et al. (2021) are given, respectively, by

$$\begin{aligned} \text{bOF}^{(i)}(t) &:= \begin{cases} v_b^{(i)}(t), & \text{if } p_b^{(i)}(t) > p_b^{(i)}(t-1), \\ v_b^{(i)}(t) - v_b^{(i)}(t-1), & \text{if } p_b^{(i)}(t) = p_b^{(i)}(t-1), \\ -v_b^{(i)}(t), & \text{if } p_b^{(i)}(t) < p_b^{(i)}(t-1), \end{cases} \\ \text{aOF}^{(i)}(t) &:= \begin{cases} -v_a^{(i)}(t), & \text{if } p_a^{(i)}(t) > p_a^{(i)}(t-1), \\ v_a^{(i)}(t) - v_a^{(i)}(t-1), & \text{if } p_a^{(i)}(t) = p_a^{(i)}(t-1), \\ v_a^{(i)}(t), & \text{if } p_a^{(i)}(t) < p_a^{(i)}(t-1). \end{cases} \end{aligned} \quad (2.3)$$

The result is a modest generalization of the order flow imbalance (OFI) proposed by Cont et al. (2014). The superior explanatory power of the OFI for market impact shown in Cont et al. (2021) provides evidence that order flow is highly correlated with future returns.

2.3. Time-insensitive set

Now, rather than combining volumes and prices, we formulate indicators separately from prices and volumes. We design the time-insensitive features depending on the data at the same tick time t (conversely, if we use data across time, we obtain a time-sensitive feature set, as we discuss below). These handcrafted features are used in Kercheval & Zhang (2015); Nousi et al. (2019).

The time-insensitive features describing the spread, mid-price, and price differences between adjacent levels of both sides are extracted. We denote the bid-ask spread for every level i as $p_a^{(i)}(t) - p_b^{(i)}(t)$, and the mid-price as $(p_a^{(i)}(t) + p_b^{(i)}(t))/2$. They form the vector

$$A := \left[p_a^{(i)}(t) - p_b^{(i)}(t), (p_a^{(i)}(t) + p_b^{(i)}(t))/2 \right]_{i=1}^n.$$

Price differences between adjacent levels are given by

$$B := \left[\left| p_a^{(i+1)}(t) - p_a^{(i)}(t) \right|, \left| p_b^{(i+1)}(t) - p_b^{(i)}(t) \right| \right]_{i=1}^{n-1} \in \mathbb{R}^{2n-2}.$$

All the features are combined to obtain the following time-insensitive set:

$$\mathbf{x}_{3t} := (A, B) \in \mathbb{R}^{4n-2}. \quad (2.4)$$

This static feature set best characterizes the price relationship on the panel.

2.4. Time-sensitive set

The time-sensitive set mainly concerns price and volume derivatives for each level i , which are defined as

$$\left[dp_a^{(i)}/dt, dp_b^{(i)}/dt, dv_a^{(i)}/dt, dv_b^{(i)}/dt \right].$$

The average time derivatives of price and volume can be computed numerically over several contiguous time ticks. The number of ticks used depends on the time interval between two ticks, which varies greatly between markets. For the Chinese A-share market, the subject of our empirical study, the time interval is usually around 3 seconds, which is much longer than the case in other more mature markets. Prices and volumes of most stocks tend to present a few changes during a 3-second time interval. Therefore, using the two most recent ticks, at time t and $t - 1$, the time-sensitive set is numerically calculated by

$$\mathbf{x}_{4t} := \left[p_a^{(i)}(t) - p_a^{(i)}(t-1), p_b^{(i)}(t) - p_b^{(i)}(t-1), \right. \\ \left. v_a^{(i)}(t) - v_a^{(i)}(t-1), v_b^{(i)}(t) - v_b^{(i)}(t-1) \right]_{i=1}^n \in \mathbb{R}^{4n}. \quad (2.5)$$

2.5. Integration of features

We combine the four feature sets defined in (2.1)-(2.5) to create a much larger feature set $\mathbf{X}_t$ that includes all of the features in $\mathbf{x}_{1t}$, $\mathbf{x}_{2t}$, $\mathbf{x}_{3t}$, and $\mathbf{x}_{4t}$. Our combination of different feature sets is inspired by the brain's structure and function. The brain is divided into several functional sections, with cooperation and communication between the different regions enabling various kinds of functions. The central nervous system integrates the received information and sends the processed signals to all parts of the body. Our LOB trades are similar operations that generate trading signals. Figure 1 shows the analogy between the brain system and the LOB trading system.

The objective of LOB trading is to classify the trading signals as *long*, *short*, or *none*. *long* means that we expect an upward trend of the stock price and buying before selling, whereas *short* indicates the opposite, selling before buying. If the stock is too stable for a trend to be captured, we denote it as the third type of trading signal, *none*. The signals are determined by

the approximate wealth changes during a fixed and limited holding period, during which we set stop-loss and take-profit points to control risks. These labels closely reflect actual transactions and earnings, with details given in Appendix A.

Although the regions of the brain work together as a network to process information for determining actions, there should be different critical areas responsible for different specific actions. Learning the relevance of the regions with functions can help us better understand how the brain, and, analogously, the LOB trading system, works. We want to determine which regions are the most important or the most irrelevant. Therefore, we want to explore the relevance of the four major features to the trading signal predictions instead of simply pursuing the accuracy of forecasts. This kind of hidden information explains the logic of black boxes and enables us to make better use of those features later.

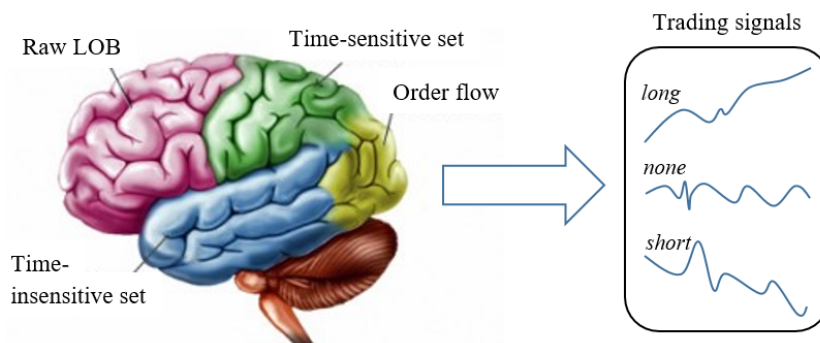


Figure 1: Illustration of how forecasting trading signals using features is analogous to the brain structure and functions. The invisible mechanism from input features to output signals needs to be explored and explained.

Another inspiration for feature testing comes from the image processing procedure under convolutional neural networks. The input features are converted into image pixels, as in Figure 2. Usually, a full picture is not required to make a judgment, and a small region is enough to make a decision. Different regions of the picture carry different levels of information, generating a ranking of feature relevance. The images composed of the feature sets enable us to explore the key points of those visualized inputs.

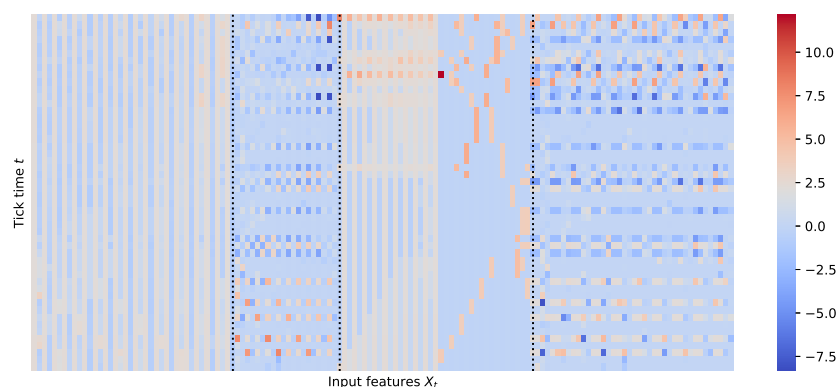


Figure 2: Heatmap of input features for 50 time ticks. The black dotted vertical lines segment the picture into four regions representing, from left to right, the feature sets $\mathbf{x}_1$, $\mathbf{x}_2$, $\mathbf{x}_3$, and $\mathbf{x}_4$. The heatmap is plotted from the normalized real data of stock SZ300052.

3. Models

Many models are designed to address multi-classification problems such as the issue that we consider. In this section, we consider four such models, ranging from naive to sophisticated, covering different levels of complexity, and including both white- and black-box models.

3.1. Multinomial logistic regression

Logistic regression is a classic statistical model that, in its basic form, uses a logistic function to model a binary dependent variable for solving classification problems. It can be extended to a multinomial type with M classes. The response $\mathbf{Y}$ containing M categorical variables is defined as

$$\mathbf{Y}_t = (y_{1t}, \dots, y_{Mt})^T, \text{ where } y_{it} = \begin{cases} 1, & \text{if } i \text{ is the true label,} \\ 0, & \text{o.w.} \end{cases} \quad (3.1)$$

The probability of being the i -th category for sample t , p_{it} , is modeled by a linear predictor $\beta_i \mathbf{X}_t$ in the following way:

$$\text{logit}(p_{it}) = \ln\left(\frac{p_{it}}{p_{Mt}}\right) = \beta_i \mathbf{X}_t, \quad (3.2)$$

where $p_{it} = P(y_{it} = 1, y_{jt} = 0, j \neq i)$. For samples with t ranging from 1 to T , the likelihood L is available and the negative log-likelihood is given by

$$L_{\log} = -\ln(L) = -\sum_{t=1}^T \sum_{i=1}^M y_{it} \log \hat{p}_{it}. \quad (3.3)$$

Because the number of parameters is very large, we add an L_1 -regularization to avoid overfitting and obtain sparse estimations by solving the following:

$$\min_{\beta} L_{\log} + \lambda \sum_{i=1}^M \|\beta_i\|_1, \quad (3.4)$$

where $\|\cdot\|_1$ denotes L_1 -Norm. The penalty is the same as the one used in LASSO regression (Tibshirani, 1996).

3.2. Multi-class support vector machine

The basic SVM (Cortes & Vapnik, 1995) is a common type of binary classifier. Given the feature $\mathbf{X}_t$ and the label $y_t \in \{1, -1\}$, the problem is formed as

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{t=1}^T \xi_t, \\ \text{subject to} \quad & y_t (\mathbf{w}^T \phi(\mathbf{X}_t) + b) \geq 1 - \xi_t, \\ & \xi_t \geq 0, \quad t = 1, \dots, T, \end{aligned} \quad (3.5)$$

where ϕ is the feature mapping function and $C > 0$ is the regularization parameter. $\phi(\mathbf{X}_t)$ maps the feature into a higher-dimensional space to allow the data to be linearly separable. The regularization is to penalize data on the wrong side of the hyperplane. Because we have a large number of features and training samples, the linear kernel function is much faster and less prone to overfitting than a nonlinear kernel. Furthermore, SVM with a linear kernel function is usually comparable with logistic regression.

For a multi-class classification model, the SVM is reduced to a set of binary classification tasks by the one-over-the-rest method and each classifier is trained independently. In the case of three classes, we need to construct three binary classifiers and make decisions based on all of the outcomes.

3.3. Random forest

The random forest method proposed by Breiman (2001) is a combination of tree predictors such that each tree depends on the values of a random vector sampled independently, and all of the trees in the forest have the same distribution. In other words, this classifier consists of a collection of tree-structured classifiers $\{h(\mathbf{X}, \Theta_k), k = 1, \dots\}$, where the $\{\Theta_k\}$ are independent and identically distributed random vectors and each tree casts a unit vote for the most popular class at input $\mathbf{X}$.

Thus, random forest is basically an ensemble learning method through majority voting based on all of the outputs from a multitude of decision trees. Therefore, it avoids the overfitting problem of an individual decision tree and is relatively stable. Generally, random forest is more advanced and captures more of the nonlinear structure than logistic regression or SVM.

3.4. Deep convolutional neural network

Combining historical LOB data, e.g., 50 ticks, the two-dimensional input resembles Figure 2, with time indicated on one axis and features indicated on the other. It is reasonable to apply convolutional layers with different shaped filters, which are designed to summarize information from each level, from the bid and ask sides, or from the nearest several time steps. Zhang et al. (2019) propose DCNN for raw LOB data that correspond to our feature set $\mathbf{x}_1$. Adopting similar DCNN models under a novel integrated LOB trading system, the prediction accuracy and actual profits are guaranteed, as shown in Yin & Wong (2022). We adapt this DCNN to deal with other feature sets, which we denote as DCNN1–DCNN4, and concat or merge them together to reach the final output. The model structure of this DCNN is summarized in Figure 3.

To reduce overfit, the dropout layer offers a very computationally cheap and remarkably effective regularization method by randomly dropping out nodes during training (Srivastava et al., 2014). Nonetheless, the number of parameters is much larger and the structure is much more complex than those of the aforementioned methods, making it hard to explain why the output is reached or which feature plays a key role. There are other interesting questions, such as whether the nonlinearity is captured and whether the features are well utilized.

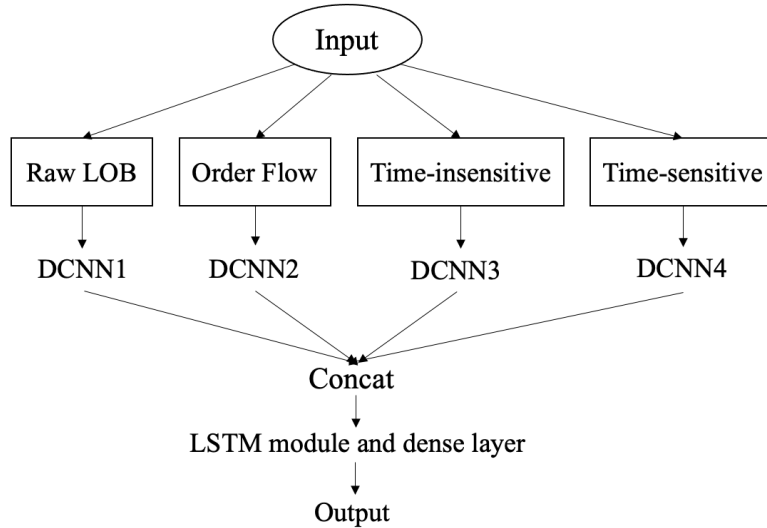


Figure 3: Model structure of the DCNN designed for learning all four feature sets. As the architecture of the individual DCNN submodels is similar, we provide details of the submodel DCNN1 only along with the hyperparameters in Appendix B.

4. Significance test of the feature sets with measures

4.1. *p*-value

In response to the limitations of the model-free hypothesis testing proposed by Candes et al. (2018) and Tansey et al. (2018), Dai et al. (2022) provide a valid *p*-value for any pre-specified region (feature) based on a black-box model. Given its wide range of applications, we expect that this will enable us to learn the relevance of the features of our DCNN model and compare our model with other accessible models. Under the model-free hypothesis, it is possible to rank the importance of features by *p*-values for all models; that is, the method is not limited to classic statistical models, traditional ML models, or deep learning models. In addition, the ability of the model to deal with different features is beneficial to further enhancing the model performance.

First, we clarify the notations with definitions. We determine a model to predict the response $\mathbf{Y}$ based on the features $\mathbf{X}$, and the loss function $l(f(\mathbf{X}), \mathbf{Y})$ calculates the discrepancy between the true value and the prediction. To capture the subtle differences between models and subsequently carry out accurate AUC score calculations, we predict the probability of each category and denote it as vector $\hat{\mathbf{P}}$. The model expressed as a function f can be learned by traditional statistical models or ML models, such as those introduced in Section 3. $R(f) = \mathbb{E}(l(f(\mathbf{X}), \mathbf{Y}))$ is the corresponding risk.

The objective is to test the relevance of a subset of features $\mathbf{X}_{\mathcal{S}} = \{\mathbf{x}_j : j \in \mathcal{S}\}$ to the prediction of $\mathbf{Y}$ with an unknown form of the prediction function, where $\mathcal{S}$ is an index set of hypothesized features and the remaining features are denoted as $\mathbf{X}_{\mathcal{S}^c} = \{\mathbf{x}_j : j \notin \mathcal{S}\}$, with $\mathcal{S}^c$ indicating the complement set of $\mathcal{S}$. Then, we replace the values of $\mathbf{X}_{\mathcal{S}}$ with zero and obtain the masked feature $\mathbf{Z}$. This comparative model is given as the prediction function g , and the risk is $R_{\mathcal{S}}(g) = \mathbb{E}(l(g(\mathbf{Z}), \mathbf{Y}))$.

For any loss function, conditional independence between the response and tested features implies risk invariance, which is expressed as

$$\mathbf{Y} \perp \mathbf{X}_{\mathcal{S}} \mid \mathbf{X}_{\mathcal{S}^c} \implies R(f^*) - R_{\mathcal{S}}(g^*) = 0, \quad (4.1)$$

where $f^* = \arg \min_f R(f)$ and $g^* = \arg \min_g R_{\mathcal{S}}(g)$. Then, the significance test for the feature set $\mathcal{S}$ is constructed in the following way:

$$H_0 : R(f^*) - R_{\mathcal{S}}(g^*) = 0, \quad \text{versus} \quad H_1 : R(f^*) - R_{\mathcal{S}}(g^*) < 0. \quad (4.2)$$

Referring to the one-split test in Dai et al. (2022), we split the given dataset $(\mathbf{X}_t, \mathbf{Y}_t)_{t=1}^T$ into the training set $(\mathbf{X}_t, \mathbf{Y}_t)_{t=1}^n$ and the testing set $(\mathbf{X}_t, \mathbf{Y}_t)_{t=n+1}^{n+m}$, with $m = T - n$. The training set includes the validation set for early stopping to prevent the overfitting problem. The trained models are expressed by the functions $(\hat{f}, \hat{g})$ to approximate (f^*, g^*) . Then, we plug these estimators into the evaluation of the testing samples. We can obtain a good estimation of $R(f^*) - R_{\mathcal{S}}(g^*)$ by the empirical risk function, which is $\hat{R}(\hat{f}) - \hat{R}_{\mathcal{S}}(\hat{g})$. As an averaged loss, the empirical risk is calculated as $\hat{R}(\hat{f}) = \frac{1}{m} \sum_{j=1}^m l(\hat{f}(\mathbf{X}_{n+j}), \mathbf{Y}_{n+j})$ for the testing set, and is similarly defined for the comparative model $\hat{g}$.

Therefore, the test statistic is given as

$$\Lambda_n = \frac{\sum_{j=1}^m \Delta_{n,j}}{\sqrt{m\hat{\sigma}_n}}, \quad \Delta_{n,j} = l(\hat{f}(\mathbf{X}_{n+j}), \mathbf{Y}_{n+j}) - l(\hat{g}(\mathbf{Z}_{n+j}), \mathbf{Y}_{n+j}) + \rho \varepsilon_j, \quad (4.3)$$

where $\hat{\sigma}_n$ is the sample standard deviation based on the inference set $\{\Delta_{n,j}\}_{j=1}^m$, and ρ is a level of perturbation to address the vanishing standard deviation. According to the asymptotic distribution of Λ_n in Theorem 2 of Dai et al. (2022), under some conditions, $\Lambda_n \xrightarrow{d} N(0, 1)$, as $n \rightarrow \infty$. The *p*-value is calculated by

$$P = \Phi(\Lambda_n), \quad (4.4)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

In statistical hypothesis testing, there is always a significance level α , which is the probability of the study rejecting the null hypothesis. Usually, α is set to 5% or 10% to determine whether the result is statistically significant. However, it is not applicable to our empirical study in Section 5.1. The averaged p -values among large amounts of stocks are much larger than the individual values. Instead of setting a rigorous threshold, we rank these features to determine their relative significance levels. In other words, the p -value is regarded as a measure of feature relevance, similar to other additional measures that we define in the following (Section 4.2). As an interesting direction for future research, we can choose a threshold after examining the overall p -value level and select relevant features from the ranking (Huynh-Thu et al., 2012).

In addition, the proposed risk-based testing is workable for any loss function, such as constant loss, L_2 -loss, or cross-entropy loss. The exact values of Λ and P are related to the choice of loss function $l(\cdot)$. However, consistently using the identical loss function gives comparable results for ranking. In our case, the loss function is the same as the one used for training models. The focal loss function originally proposed in Lin et al. (2017) is designed for imbalanced data, and is written as

$$l(\hat{\mathbf{Y}}_t, \mathbf{Y}_t) = - \sum_{i=1}^M b_i (1 - \hat{y}_{it})^\gamma y_{it} \log \hat{y}_{it}, \quad (4.5)$$

where b_i and γ are additional hyperparameters used to adjust the class weights. The effect of the modulating factor $(1 - \hat{y}_{it})^\gamma$ is increased with γ . On the contribution of the focal loss function compared with the traditional categorical cross-entropy loss function, refer to Yin & Wong (2022).

4.2. *R-squares*

We propose additional measures, apart from the p value, to evaluate the features' relevance in a comprehensive manner. The measures are derived from the coefficient of determination and modified according to the most general definition, which is $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$. This classic statistical measure determines the proportion of variance in the dependent variable that can be explained by the independent variable. In the spirit of its interpretation, we define the out-of-sample $\tilde{R}^2$ to measure the explanatory power of the hypothesized feature $\mathcal{S}$, and we generalize it in terms of the loss function and AUC score. A similar self-defined out-of-sample R^2 in Gu et al. (2020) is used to assess predictive performance for individual excess stock return forecasts, and is further applied in Leippold et al. (2022) to determine the most important predictors in the Chinese and US markets.

Instead of using the most common mean squared error as in the definition of R^2 , we quantify the model error by the loss functions. The negative consequences of error can be measured by symmetric or asymmetric loss functions. Although the loss function is widely applied in regression or classification problems, calculating the AUC score is a very popular means of measuring the ability of a classifier to distinguish between classes (Hanley & McNeil, 1983). Huang & Ling (2005) show theoretically and empirically that the AUC is preferable to total accuracy as a classifier evaluation. The AUC score is desirable and useful because it is invariant to scale and classification thresholds, and is coherent, especially for imbalanced data (Flach et al., 2011; He & Garcia, 2009).

As we mention above, the binary classifier AUC stands for “the area under the curve,” where the “curve” refers to the receiver operating characteristic (ROC) curve, which plots the true positive rate against the false positive rate at various threshold settings. That means we need to input the predicted probabilities instead of labels to obtain an accurate ROC curve. AUC ranges in value from 0 to 1; the higher (lower) the AUC, the better (worse) is the performance of the model in distinguishing positive and negative classes. We can easily extend it to multi-class classification problems by using the one-over-the-rest technique and a micro-average that will

aggregate the contributions of all classes to compute the average metric. Considering that our dataset is imbalanced, the micro-average is preferable because it assigns unequal weights to the labels.

In the context of a predictive or inferential model, we evaluate the performance of each subset of features $\mathcal{S}$ by using the out-of-sample $\tilde{R}^2$ for each testing period, defined as

$$\tilde{R}_{\text{error}}^2 := \sum_{j=1}^m l(\hat{g}(\mathbf{Z}_{n+j}), \mathbf{Y}_{n+j}) / \sum_{j=1}^m l(\hat{f}(\mathbf{X}_{n+j}), \mathbf{Y}_{n+j}) - 1, \quad (4.6)$$

$$\tilde{R}_{\text{AUC}}^2 := 1 - s\left(\left[\hat{g}(\mathbf{Z}_{n+j}), \mathbf{Y}_{n+j}\right]_{j=1}^m\right) / s\left(\left[\hat{f}(\mathbf{X}_{n+j}), \mathbf{Y}_{n+j}\right]_{i=1}^m\right), \quad (4.7)$$

where $l(\cdot)$ is the loss function and $s(\cdot)$ is the AUC score. Other notations are the same as in Section 4.1 for calculating the p -value. Generally, the larger (smaller) the $\tilde{R}_{\text{error}}^2/\tilde{R}_{\text{AUC}}^2$, the more (less) significant is the feature set $\mathcal{S}$. The values of $\tilde{R}_{\text{error}}^2/\tilde{R}_{\text{AUC}}^2$ might be negative in some cases where the masked features $\mathbf{Z}$ are better than the original whole feature set $\mathbf{X}$ for prediction. Tested features are not significant and not contributing, merely providing noise or other misleading information.

One should be cautious about comparing these two measures between different models. Because they are highly correlated with the original model performance and are calculated based on the improvement or degradation brought about by the selected features, the values are model-dependent. To some extent, the ranking of hypothesized features is usually restricted to testing under an identical initial model. Cross-model comparisons of absolute $\tilde{R}^2$ values require consideration of additional scaling or regularization.

5. Results of the feature relevance

5.1. Empirical study on the Chinese A-share market

We study the marginal effect of the individual feature sets and the joint importance of the combined feature sets. The index set of hypothesized features is denoted as

$$\mathcal{S} \in \{1, 2, 3, 4, 12, 13, 14, 23, 24, 34, 123, 124, 134, 234\},$$

where 1, 2, 3, 4 represent the raw LOB $\mathbf{x}_1$, the order flow $\mathbf{x}_2$, the time-insensitive set $\mathbf{x}_3$, and the time-sensitive set $\mathbf{x}_4$, respectively. Other numbers represent combinations of the specified feature sets; for example, 12 is the joint set $\{\mathbf{x}_1, \mathbf{x}_2\}$ and 123 combines the feature sets $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$.

The 10-level LOB data used for the experiments are level-II tick data from two major Chinese A-share markets, the Shanghai Stock Exchange (SHSE) and the Shenzhen Stock Exchange (SZSE). The proprietary data are provided by the Fintech company TradeMaster¹. Because of the computational time and data availability, we empirically analyze the historical LOB data covering 30 stocks and five periods. The total testing period for one stock lasts over two months (50 weekdays). Models are trained at the individual stock level, and features are tested under a total of 150 sets of data. We use the four models in Section 3 in the empirical study, denoted by their abbreviations. We use box and bar plots to display summary statistics, as shown below, and the results of the same model are represented in the same color in the figures.

First, Figure 4 shows the p -values of all of the hypothesized features $\mathcal{S}$ under two models, the classic logistic regression (LR) model and the advanced DCNN model. The white hollow dots represent the average of individual stock results, and the boxplot displays the distribution of p -values in regard to stocks. We find that there are huge disparities in the magnitude of p -values

¹<https://www.trademastertech.com>

between stocks. However, the rank of feature importance is much more stable and less sensitive to stock selection, especially from a stock pool perspective, which is discussed in Section 5.3. Moreover, estimated p -values for features tested by the DCNN model are much smaller than those produced by the LR model. For such a high-dimensional intricate problem with noisy and non-stationary characteristics, deep learning techniques are considered to provide stronger support for feature engineering than other more preliminary models (Zhu et al., 2018; Long et al., 2019).

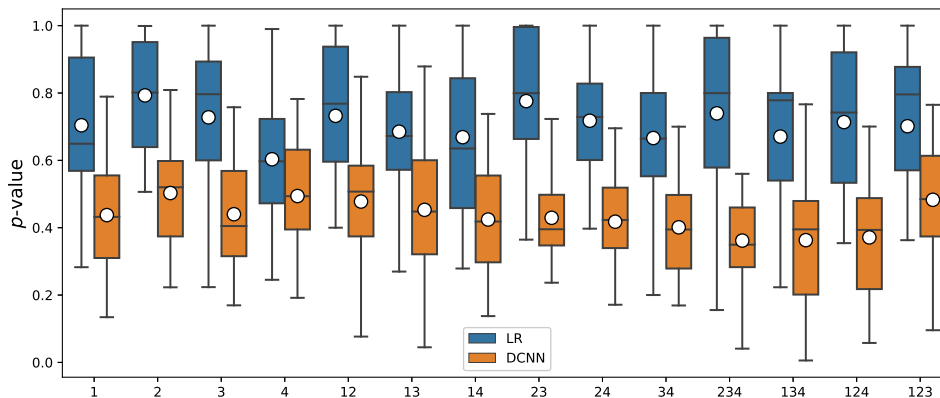


Figure 4: Boxplot of p -value estimated for hypothesized features under the logistic regression and DCNN. The white hollow dots refer to the mean and the black horizontal lines refer to the median.

From a stock pool perspective, we take the average of all stock results and use the obtained values for the significance ranking of the hypothesized features $\mathcal{S}$. The estimated values of three measures (p -value, $\tilde{R}_{\text{error}}^2$, and $\tilde{R}_{\text{AUC}}^2$) for the four models (LR, SVM, RF, and DCNN) are presented in Figure 5-6. The order of the features for the last two measures ($\tilde{R}_{\text{error}}^2$ and $\tilde{R}_{\text{AUC}}^2$) are identical to the order when the features are sorted by p -value, where the significance decreases from top to bottom in the subplots of the first column. This is for convenience and consistency in comparing the measures. As for the magnitude of $\tilde{R}_{\text{error}}^2$ and $\tilde{R}_{\text{AUC}}^2$, negative values identify which features degrade the model's performance over time, and positive values monitor how much the features contribute.

We illustrate the findings from two aspects, namely the consistency between the three measures and the significance of the hypothesized features. For the LR, it is obvious that the three measures do not agree with each other, as shown by the messy shapes of the bars in Figure 5. For SVM, the same feature set is indicated as most significant by all of the measures, but most of the features are not significant, as revealed by a negative out-of-sample $\tilde{R}^2$ and the p -values being larger than 0.5. For random forest, we discover a clear gap between outperforming and underperforming features, which is consistent for the three measures. There is a great discrepancy between the good and the bad, further demonstrating that the random forest has distinct preferences for features. For DCNN, both the consistency of measures and the significance of features perform well to a certain extent. In line with the order obtained by the p -values, the decreasing characteristics from top to bottom of the out-of-sample $\tilde{R}^2$ are evident. We can identify several important feature sets through small p -values and large positive values for $\tilde{R}_{\text{error}}^2$ and $\tilde{R}_{\text{AUC}}^2$. The importance of three combined feature sets ($\{234, 134, 124\}$) is particularly prominent under DCNN, probably because this model has excellent processing ability for high-dimensional data.

The aim of the above analysis is to explore the meaning of features relative to the same original model and to rank them according to the measures. This comparative testing does not strictly reveal the predictive performance of these models. Therefore, we examine the original models and perform cross-model comparisons of the prediction error, AUC scores, and signal profits after trading. The profits are obtained by taking the signals for investment with the

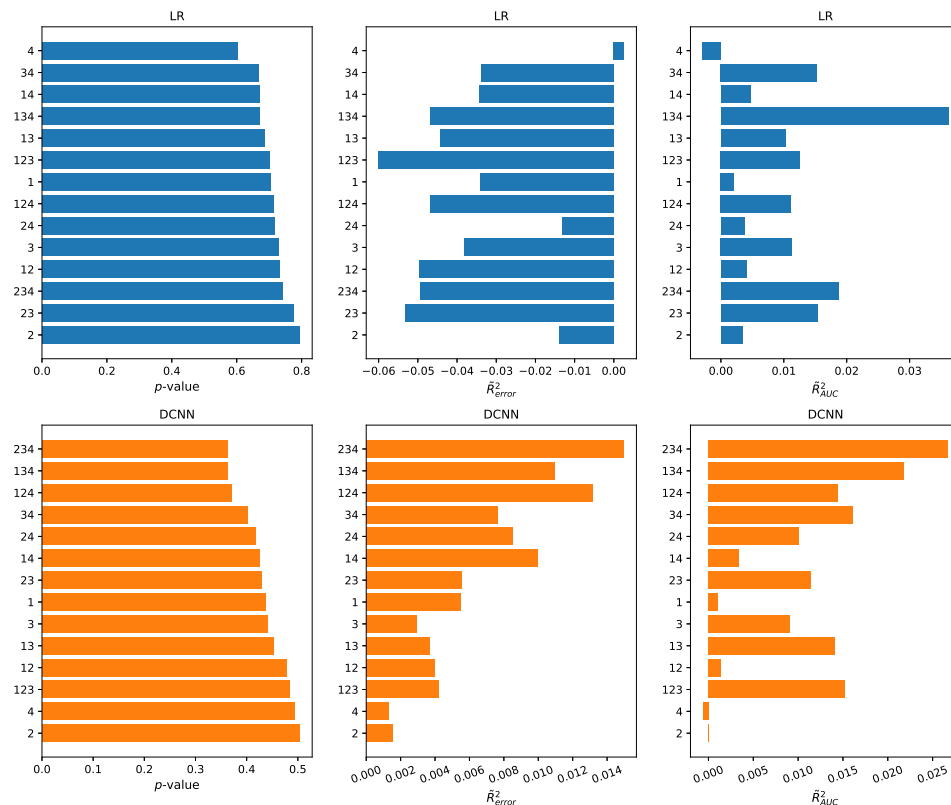


Figure 5: Barplot of the three measures estimated for the hypothesized features under the logistic regression and DCNN. The reference ranks are based on the left panels, labeled “ p -value.”

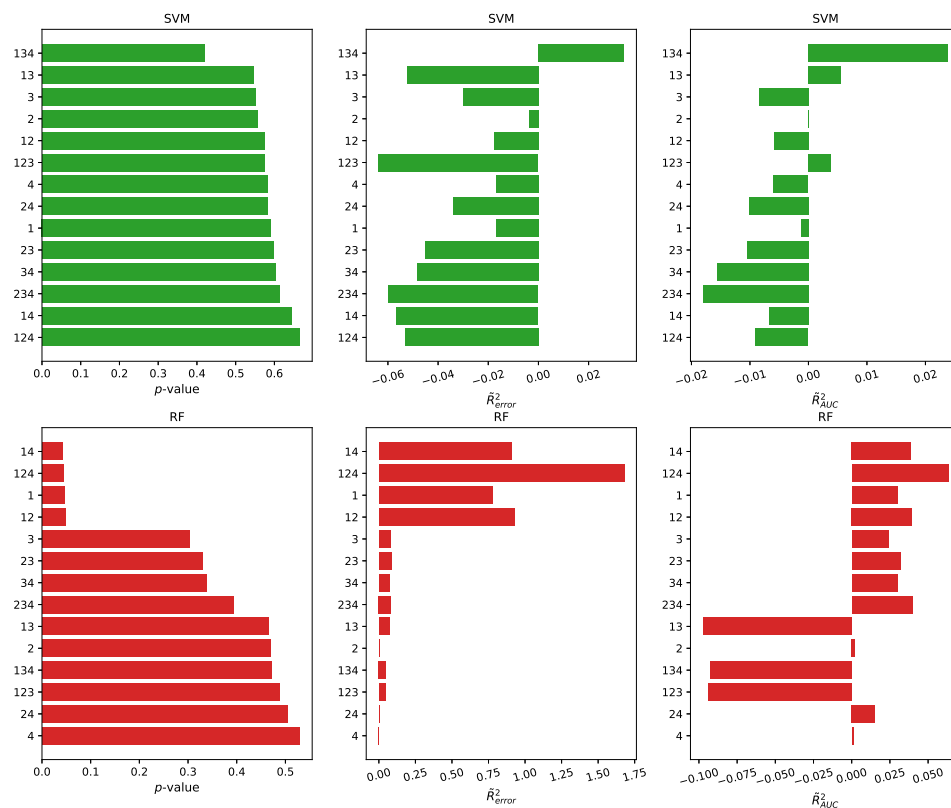


Figure 6: Barplot of three measures estimated for the hypothesized features under traditional ML models (SVM and random forest). The reference ranks are based on the left panels, labeled “ p -value.”

same tick-by-tick trading strategy proposed in Yin & Wong (2022), with details provided in Appendix C. The unit of profit is basis points (bps). In contrast to the four models that utilize features, the benchmark here is to estimate the signal probability in the absence of feature information and count the ratio of each class in the training set as the predicted probability. All of the models are tested under the same stock pool, 30 stocks, using one-pool training, and the comparison of models is shown in Table 1.

	Benchmark	LR	SVM	RF	DCNN
Mean of error	0.9123	0.8863	0.5151	0.7084	0.4508
Mean of AUC score	0.5948	0.6632	0.6497	0.6738	0.7621
Average profit per signal (bps)	-	15.63	10.10	20.86	27.43
Std of profit	-	0.0055	0.0044	0.0049	0.0046

Table 1: Performance of original models in terms of prediction error, AUC score, and signal profit after trading.

Our DCNN outperforms the other models in almost every aspect. The random forest is the second-best model, as it is more satisfactory than LR and SVM. It is not difficult to infer that measures such as the Sharpe ratio support the preference for DCNN. Table 1 shows the significant differences in the ability of the models to process features and extract information. Part of our results can be supported by the experimental comparisons and conclusions in Brown & Mues (2012); Shin & Balasingham (2017); Couronné et al. (2018), where similar algorithms are applied to classify various real data sets. In addition, Table 1 provides evidence for taking the consistency of the measures and the significance of the features as references for the original model performance. Although the feature testing and importance results are model-dependent, they indicate the cross-model preference based on the shapes of the barplots presented in Figure 5-6. By measuring the feature relevance, we identify hidden patterns, including consistency and significance, that provide explanations and support for the choice of models.

5.2. Ensemble learning

From the above results, we find that the feature importance or relevance varies greatly in different models and measures. Based on the p -value, $\tilde{R}_{\text{error}}^2$, and $\tilde{R}_{\text{AUC}}^2$, we determine the consistently most significant feature sets. Therefore, we consider an ensemble DCNN with features from $\{2, 3, 4\}$ and the random forest model trained by feature set $\{1, 2, 4\}$. It is intended to handle the specific feature sets separately. Originally, ensemble systems are developed to reduce the variability in classifier decisions and thereby increase the generalization performance, including data sampling/selection, training member classifiers, and combining classifiers (Dietterich, 2000; Polikar, 2012). In particular, ensemble learning can be applied to handle imbalanced data and its generative model turns out to be superior to the separate models (Sun et al., 2015; Yijing et al., 2016). We adopt the idea of the ensemble to exploit the significant features of each model as an additional means of improving the predictive power.

One of the earliest and simplest yet effective ensemble-based algorithms, the bagging mechanism, is applied to train the DCNN and the random forest classifiers independently. However, instead of simple majority voting or averaging, the weights for the ensemble combination are chosen by the model performance based on validation data ($t = n_0, \dots, n$) within the training set ($t = 1, \dots, n$). Admittedly, this random forest model is preliminary and without advanced settings, and its predictive power is not very stable. Therefore, we choose a quite small weight, ω , for the ensemble in this study. Although the contribution of the random forest seems mild compared to the DCNN model, implementing these signals improves the prediction accuracy in terms of AUC score as well as the actual profit, as shown in Table 2.

The trained models by RF and DCNN are expressed by the functions ($\hat{g}_{\text{RF}}$ and $\hat{g}_{\text{DCNN}}$), respectively, and the selected features from $\mathcal{S}$ are denoted as $\mathbf{X}^{\mathcal{S}}$. Unlike the masked feature

$\mathbf{Z}$, which replaces the values with zero, we drop these useless variables and reduce the model to fit the selected feature dimension. Then, by a specific ensemble rule, the predicted probabilities for the inference set are given as

$$\hat{\mathbf{P}}_{n+j} = \omega_n * \hat{g}_{\text{RF}}(\mathbf{X}_{n+j}^{124}) + (1 - \omega_n) * \hat{g}_{\text{DCNN}}(\mathbf{X}_{n+j}^{234}), \quad j = 1, 2, \dots, m, \quad (5.1)$$

where $\omega_n = \arg \max_{\omega \in \Omega} s([\omega * \hat{g}_{\text{RF}} + (1 - \omega) * \hat{g}_{\text{DCNN}}, \mathbf{Y}_t]_{t=n_0}^n)$ depending on the validation data, $\Omega = [0, 1]$, and $s(\cdot)$ is the AUC score function introduced before.

	Original DCNN	Original RF	Original ensemble model
Mean of error	0.4508	0.7084	0.4826
Mean of AUC score	0.7621	0.6738	0.7539
Average profit per signal (bps)	27.43	20.86	29.01
Std of profit	0.0046	0.0049	0.0047
	DCNN ²³⁴	RF ¹²⁴	Ensemble selected model
Mean of error	0.4336	0.7240	0.4424
Mean of AUC score	0.7822	0.6876	0.7825
Average profit per signal (bps)	29.87	22.21	32.98
Std of profit	0.0039	0.0037	0.0040

Table 2: Performance in terms of prediction error, AUC score, and signal profit after trading for different models. The original DCNN and RF models are those trained using all features. DCNN²³⁴ and RF¹²⁴ are trained using selected features.

To fit the selected features $\mathbf{X}^S$, the models are reduced to a simpler structure with fewer parameters and lower dimensions than the original models. Such refinements enhance the model performance compared with the original models trained using the full feature sets. Combining them again afterward avoids the omission of information and seeks to improve payback. The outcomes in Table 2 indicate the improvements yielded by proper feature selection and model ensemble techniques. The experiments in this section provide enlightening ideas on how to leverage the feature testing results to improve performance, especially for black-box models.

5.3. Sensitivity analysis

In previous sections, we investigated the measure consistency and feature significance for four models. However, some settings and operations in the empirical study may influence the results; to distinguish them from model parameters and machine hyperparameters, we refer to them as off-machine parameters. Specifically, as a sensitivity analysis, we conduct experiments for three off-machine parameters: the time period, data split, and stock selection. The sensitivity analysis for the three off-machine parameters tests the robustness of the rankings determined by the measures and helps to confirm their applicability. Because of space limitations and our focus on black-box models, we only test the DCNN model. Thus, the following results all relate to the DCNN.

In Section 5.1 above, we test five periods for each stock and then take the average of the results for all stocks and all periods without distinguishing the specific time period. These results are considered to represent the overall performance throughout the whole testing period. In Figure 7, we examine the results calculated from each period separately to check the consistency between neighboring time periods. The reference ranks are taken from the third period, indicated by the orange bars in the middle panel of Figure 7. We maintain the feature order of this reference group for other periods for clarity. After a horizontal comparison, we conclude that with slight and acceptable performance differences in several features, the ranking of feature importance exhibits sufficient stability and consistency over the nearby time periods.

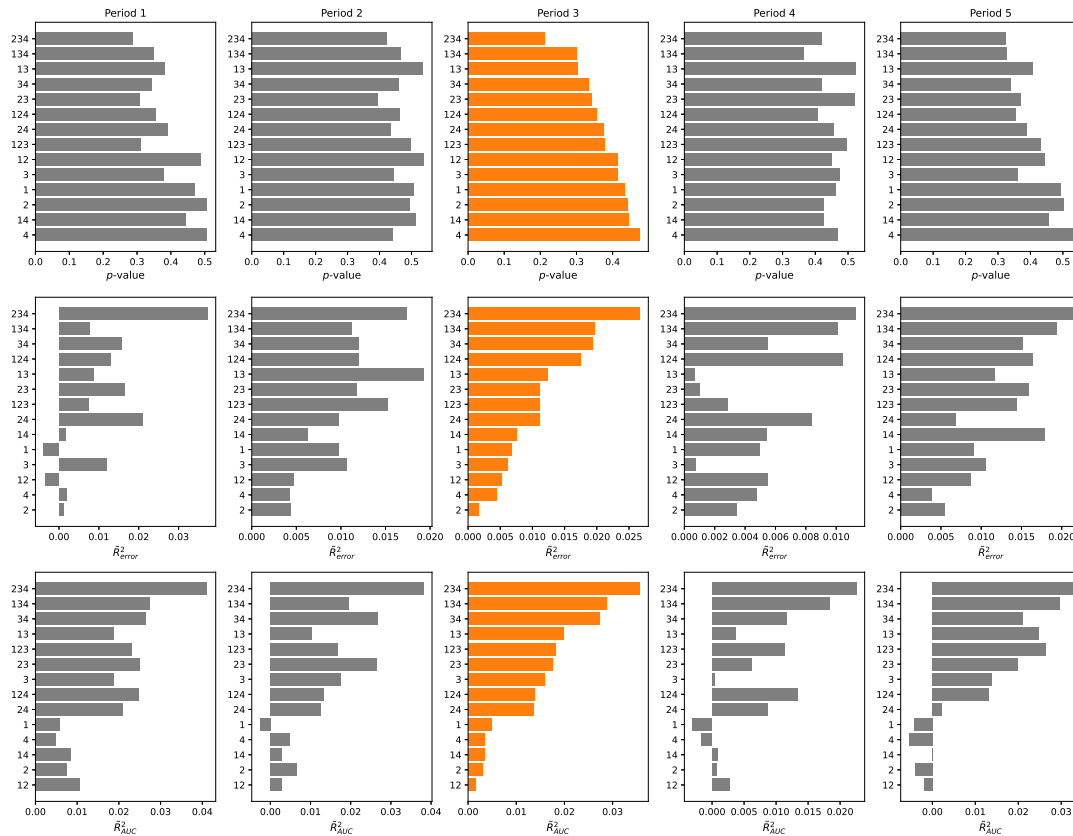


Figure 7: Barplot of three measures estimated for hypothesized features under DCNN over five periods. The reference ranks are from the middle orange panels, labeled “period 3.”

The second analysis is related to the data split. Instead of dividing the data into five periods as in Section 5.1, what will happen if we simply test the entire long period? Will the ranking be greatly affected? To answer these questions, we estimate the values of the three measures over the whole period, shown by the gray bars in Figure 7, and compare them with the previous results, indicated by the orange bars in Figure 5. The results, shown in Figure 8, suggest that the significance test is not sensitive to the data splitting. Although the exact values of the two methods differ from each other, they give similar rankings of features.

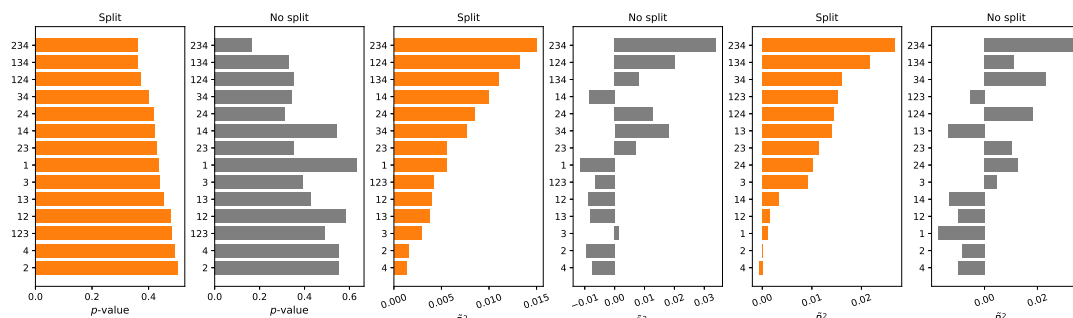


Figure 8: Barplot of three measures estimated for hypothesized features under DCNN using two calculation methods. “Split” indicates the average from testing five periods and “no split” indicates testing with the five periods combined. The reference ranks are from the orange panels, labeled “split.”

We conduct another sensitivity analysis on the effect of stock selection on feature ranking. We randomly select another 30 stocks as a pool to compare with the stock pool tested earlier in Section 5.1. All of the other settings remain the same. The feature testing indicates robustness,

as shown in Figure 9, where the two colors denote the two stock pools and the reference order is taken from the orange panels labeled “split.”

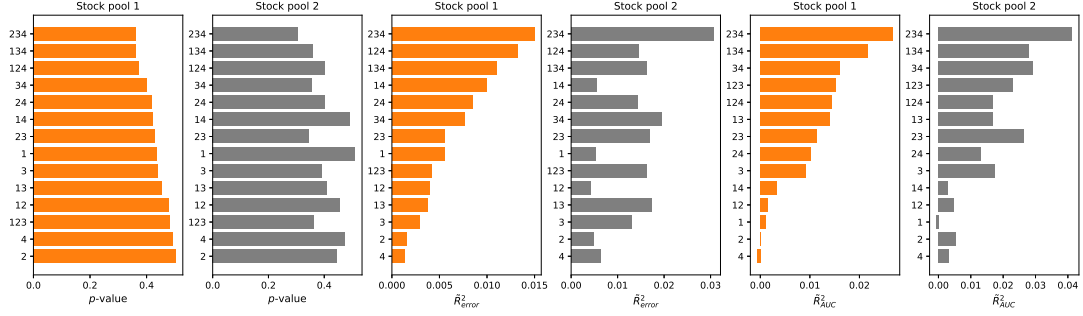


Figure 9: Barplot of three measures estimated for the hypothesized features under DCNN for two groups of stocks. “Stock pool 1” contains 30 stocks, studied in Section 5.1, and “stock pool 2” contains 30 other random stocks. The reference ranks are from the orange panels, labeled “stock pool 1.”

The experiments enhance our confidence in the results and provide support for the application of these statistics to test a vast number of features under black-box models. Off-machine parameters, including the time period, data split, and stock selection, do not greatly affect the rankings. The robustness indicates the feasibility of using the approach in a wide range of areas in practice and shows that it is reasonable to estimate these values for ranking features, especially for complex deep learning models. Moreover, the hidden black-box information revealed by the results is convincing and worth exploiting for model improvement.

6. Conclusion

This paper provides three statistical measures to test feature relevance using black-box models with an integral investigation of LOB trading. We conduct significance tests for four representative feature sets and their combinations using a classic statistical model, traditional ML models, and an advanced deep learning model, the DCNN. The empirical study of LOB trading on the Chinese A-share market indicates that the consistency of the measures and the significance of the features vary across the models and are correlated with algorithm performance. In particular, under the DCNN, features have consistent rankings by measures, and most are committed to forecasting accuracy. Extended experiments, including ensemble learning for model improvement and sensitivity analyses for off-machine parameters, emphasize the benefits of explaining features and encourage the utilization of the proposed measures.

Appendix A. Details of the labeling

The labeling for the trading actions follows Yin & Wong (2022); see Section 2.3 therein. We briefly illustrate the main idea here. The approximate wealth change for a *short* signal is given as

$$c_t^{short} \triangleq \begin{cases} 2 - \underline{\beta}, & \text{if } t < \min \underline{\mathcal{S}} < \min\{\bar{\mathcal{S}}, t + \Delta\}, \\ 2 - \bar{\alpha}, & \text{if } t < \min \bar{\mathcal{S}} < \min\{\underline{\mathcal{S}}, t + \Delta\}, \\ \frac{2p_b(t) - \min_{s \in [t, t+\Delta]} p_a(s)}{p_b(t)}, & \text{if } \underline{\mathcal{S}} = \bar{\mathcal{S}} = \emptyset, \end{cases} \quad (\text{A.1})$$

where $\underline{\mathcal{S}} := \{s \in [t, t + \Delta] : p_a(s) < \underline{\beta}p_b(t)\}$, $\bar{\mathcal{S}} := \{s \in [t, t + \Delta] : p_a(s) > \bar{\alpha}p_b(t)\}$, $\bar{\alpha}$, and $\underline{\beta}$ are hyperparameters for stopping early. Δ is the length of the limited holding time and can be set at 1, 2, or more minutes by traders. Conversely, the approximate wealth change for a *long* signal is given by

$$c_t^{long} \triangleq \begin{cases} \bar{\beta}, & \text{if } t < \min \underline{\mathcal{S}}' < \min\{\bar{\mathcal{S}}', t + \Delta\}, \\ \underline{\alpha}, & \text{if } t < \min \bar{\mathcal{S}}' < \min\{\underline{\mathcal{S}}', t + \Delta\}, \\ \frac{\max_{s \in [t, t+\Delta]} p_b(s)}{p_a(t)}, & \text{if } \underline{\mathcal{S}}' = \bar{\mathcal{S}}' = \emptyset, \end{cases} \quad (\text{A.2})$$

where $\underline{\mathcal{S}}' := \{s \in [t, t + \Delta] : p_b(s) > \bar{\beta}p_a(t)\}$ and $\bar{\mathcal{S}}' := \{s \in [t, t + \Delta] : p_b(s) < \underline{\alpha}p_a(t)\}$ with hyperparameters $\underline{\alpha}$ and $\bar{\beta}$.

Based on the approximate wealth returns for *long* and *short* signals, the labels of the three corresponding trading actions are

$$l_t = \begin{cases} +1, & \text{if } c_t^{long} > \max\{c_t^{short}, \epsilon\}, \\ -1, & \text{if } c_t^{short} > \max\{c_t^{long}, \epsilon\}, \\ 0, & \text{o.w.}, \end{cases} \quad (\text{A.3})$$

where ϵ is the threshold for checking the strength of signals. In the above equation, +1 denotes taking the long position, -1 denotes taking the short position, and 0 denotes doing nothing at the current time t . Thus, a larger (smaller) ϵ denotes a more (less) stringent signal screening. Yin & Wong (2022) provide an approach to determine the values of the hyperparameters, $\bar{\alpha}$, $\underline{\alpha}$, $\bar{\beta}$, and $\underline{\beta}$ based on historical bid-ask spreads from the training and validation datasets.

Appendix B. Details of the submodel DCNN1

Module	Layer (options)			Output shape	Number of parameters
	Input			[(None, 50, 40, 1)]	-
Convolutional layers	Conv2D (1×2@16, stride=1×2)			(None, 50, 20, 16)	48
	Conv2D (5×1@16)			(None, 50, 20, 16)	1296
	Conv2D (1×2@16, stride=1×2)			(None, 50, 10, 16)	528
	Conv2D (5×1@16)			(None, 50, 10, 16)	1296
	Conv2D (1×10@16)			(None, 50, 1, 16)	2576
	Conv2D (5×1@16)			(None, 50, 1, 16)	1296
Inception module	Conv2D (1×1@16)	Conv2D (1×1@16)	MaxPooling2D(3×1)	(None, 50, 1, 48)	2896
	Conv2D (3×1@16)	Conv2D (5×1@16)	Conv2D (1×1@16)		
	Filter concatenation				
-	Reshape and Dropout			(None, 50, 48)	0

Table B.3: Summary of the DCNN submodel (DCNN1) structure designed for learning the feature set $\mathbf{x}_1$, raw limit order book.

Details on the hyperparameters: the optimizer is the adaptive moment estimation algorithm (ADAM) with default parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$ from Kingma & Ba (2014); the batch size is 128; the learning rate is 0.01; and the activation function for Conv2D is LeakyReLU with a negative slope coefficient of $\alpha = 0.01$.

Appendix C. Details of the trading strategy

Figure C.10 illustrates the procedure for converting the signal predictions into trading actions. The trading strategy involves practical concerns and risk control.

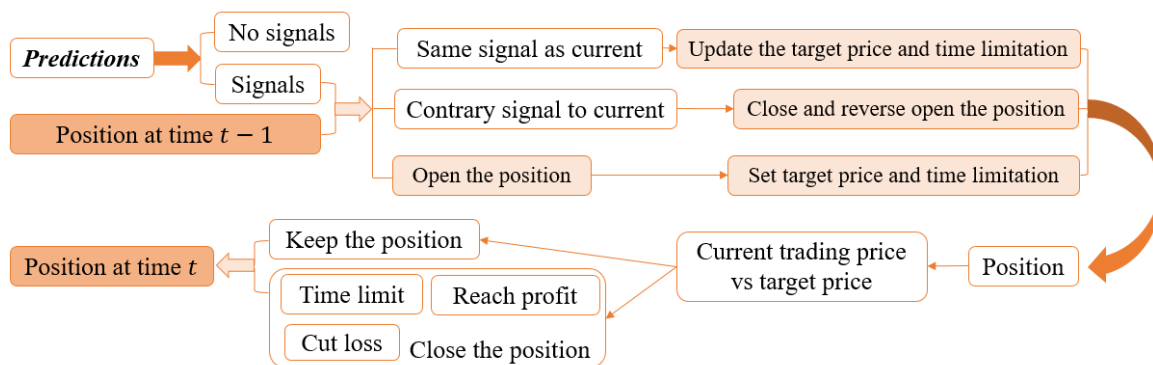


Figure C.10: The tick-by-tick trading strategy for exploiting signal predictions to update the account position.

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